Data Sciences - CS3072

Data-Driven Analysis of EEG Signals for Epileptic Seizure Detection

Research Paper

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1. Abstract

Epilepsy is a chronic neurological disorder affecting millions worldwide and is characterized by recurrent and unpredictable seizures caused by abnormal brain activity. Early detection and continuous monitoring are essential for improving patient safety and enabling timely medical intervention. Electroencephalography (EEG) is widely used to record brain activity and detect seizures.

This study analyzes EEG signal data to identify patterns that distinguish seizure activity from normal brain activity. The dataset is transformed into a binary classification problem, and machine learning models are applied to automatically detect seizures. The ultimate goal is to support the development of intelligent and real-time wearable seizure monitoring systems.

2. Introduction

Epilepsy can lead to physical injuries, psychological stress, and reduced quality of life. Early and accurate seizure monitoring is crucial, as seizures often occur unpredictably. EEG signals provide valuable insights into brain activity, but their high dimensionality and non-stationary nature make automated analysis challenging. Data-driven approaches using machine learning and deep learning offer promising solutions for developing automated seizure detection systems.

2.1 Problem Statement

Despite advances in medical technology, seizure detection remains a significant challenge. Traditional methods rely on hospital-based EEG recordings and manual interpretation by

clinicians. While these approaches are reliable in clinical environments, they have several limitations when used for continuous monitoring.

EEG monitoring in hospitals is often limited to short observation periods, meaning seizures that occur outside clinical settings may go undetected. In addition, manual analysis of EEG recordings is time-consuming and requires expert knowledge. Wearable seizure monitoring systems introduce further challenges due to patient variability, limited electrode placement, and real-time processing constraints [3].

To address these limitations, researchers have increasingly explored data-driven approaches using machine learning and deep learning techniques. These approaches can automatically learn patterns from EEG signals and support the development of automated seizure detection and monitoring systems [1][2][3].

2.2 Objective of This Study

This research focuses on analyzing EEG signals to identify patterns that distinguish seizure activity from normal brain activity. By applying preprocessing and exploratory data analysis techniques, the study aims to better understand the characteristics of EEG signals and support the development of smart wearable seizure monitoring systems.

3. Literature Review

Automated epileptic seizure detection using EEG signals has become important because traditional methods rely on manual analysis by doctors, which takes time and may lead to errors. EEG data is also large and complex, so analyzing it manually is difficult. Because of this, many researchers have started using machine learning and deep learning techniques to improve seizure detection.

In earlier studies, researchers used traditional machine learning methods that required manual feature extraction. This means experts had to select important features from EEG signals before applying a model. One important approach is the use of Long Short-Term Memory (LSTM), which is designed to learn patterns over time. Since EEG signals are time-based, LSTM models are useful for detecting seizures. Studies have shown that LSTM can successfully learn these patterns and achieve good prediction results [4].

Later, deep learning models such as Convolutional Neural Networks (CNNs) became more popular. CNNs can automatically learn features from raw EEG data, which makes them easier to use and more efficient. They are especially good at detecting patterns in data, which helps improve classification accuracy. Research shows that CNN-based models perform better than many traditional methods in seizure detection tasks [5].

To further improve results, some studies combined different models together. For example, CNN and LSTM are often used in one system to capture both spatial and temporal patterns in EEG signals. CNN extracts important features, while LSTM learns how these features change over time. These hybrid models usually perform better than using only one model because they can understand the data more deeply [6].

Other studies combined deep learning with traditional classifiers such as Support Vector Machines (SVM). These methods use deep learning for feature extraction and SVM for classification. This combination has shown high accuracy and strong performance on EEG datasets [7].

Recently, researchers have focused on improving how well models work on different patients and real-world data. EEG signals can vary from one person to another and may contain noise. To solve this, techniques like ensemble learning and transfer learning are used. These methods help make the models more reliable and accurate in real situations [2].

Another recent study proposed a deep learning framework for epileptic seizure detection using EEG signals. The researchers applied neural network models that automatically learn complex patterns from EEG data without requiring manual feature selection. Their results showed that deep learning methods can effectively detect seizure activity and achieve high accuracy when analyzing EEG recordings. This study highlights the potential of advanced neural network architectures to improve automated seizure monitoring systems and support real-time seizure detection [9].

Similarly, another research study developed a real-time seizure detection approach using EEG signals combined with signal processing techniques and convolutional neural networks. The proposed method analyzed EEG signals using time-frequency analysis and deep learning to identify seizure patterns. The study reported high detection performance and demonstrated that combining signal processing methods with deep learning models can significantly improve the reliability of automated seizure detection systems [10].

Despite the significant progress made in EEG-based seizure detection, several challenges still remain. Many existing studies focus mainly on developing complex machine learning and deep learning models, but they often require large computational resources and carefully prepared datasets. In addition, EEG signals are highly variable between patients and may contain noise or artifacts, which can reduce model accuracy in real-world scenarios. Some models also perform well in controlled experimental environments but may struggle when applied to real-time monitoring systems or wearable devices.

Therefore, understanding the structure and behavior of EEG data is an important step before developing reliable seizure detection systems. The goal of this study is to analyze EEG signals using preprocessing and exploratory data analysis techniques in order to identify patterns that distinguish seizure from non-seizure activity. By examining the characteristics of EEG data, this research aims to provide insights that can support the development of intelligent seizure monitoring systems, particularly those integrated with wearable healthcare technologies.

4. Methodology

4.1 Data Information

The dataset used in this study is the Epileptic Seizure Recognition Dataset, which contains EEG recordings used for seizure detection. The dataset is publicly available on Kaggle and is widely used for benchmarking machine learning models in biomedical signal analysis.

After preprocessing, the dataset contains 9,490 EEG signal samples. Each sample represents a short segment of brain activity recorded over time. Each instance includes 178 numerical features (X1–X178), which represent EEG signal amplitudes measured at consecutive time steps. In addition, each sample has a target label (y) indicating the type of brain activity.

The original dataset consists of five classes representing different brain states. In this study, the problem is reformulated into a binary classification task:

- Class 1 → Seizure activity
- Classes 2–5 → Non-seizure activity

This transformation allows the model to focus on distinguishing abnormal brain activity from normal brain patterns, which is essential for seizure detection systems.

4.2 Data Description

The dataset consists entirely of numerical EEG signal values that represent electrical brain activity over time. Each row corresponds to one EEG signal segment, while each column represents the signal amplitude at a specific time step. Therefore, the dataset is structured as a time-series representation rather than independent tabular features.

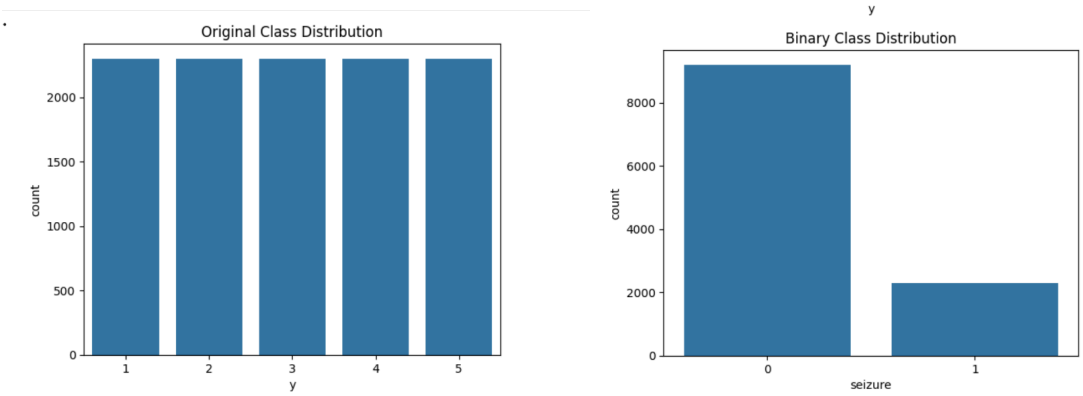
EEG signals naturally contain temporal dependencies, where patterns such as spikes, waves, and irregular fluctuations may indicate seizure activity. Seizure signals typically exhibit sudden high-amplitude spikes and chaotic variations, whereas non-seizure signals remain relatively stable.

Before analysis, several preprocessing steps were applied. The non-informative “Unnamed” column was removed, and the dataset was checked for missing values and duplicates, with none detected. The target variable was converted into a binary format for classification purposes.

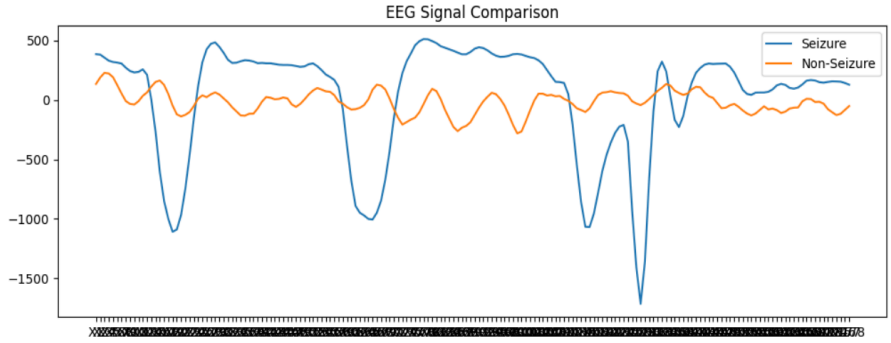
Although no explicit feature reduction or augmentation techniques were applied, the high dimensionality (178 time steps) inherently captures sufficient temporal information. Feature scaling (StandardScaler) was applied during model training to normalize feature ranges and improve the performance of distance-based models such as SVM.

4.3 Data Visualization

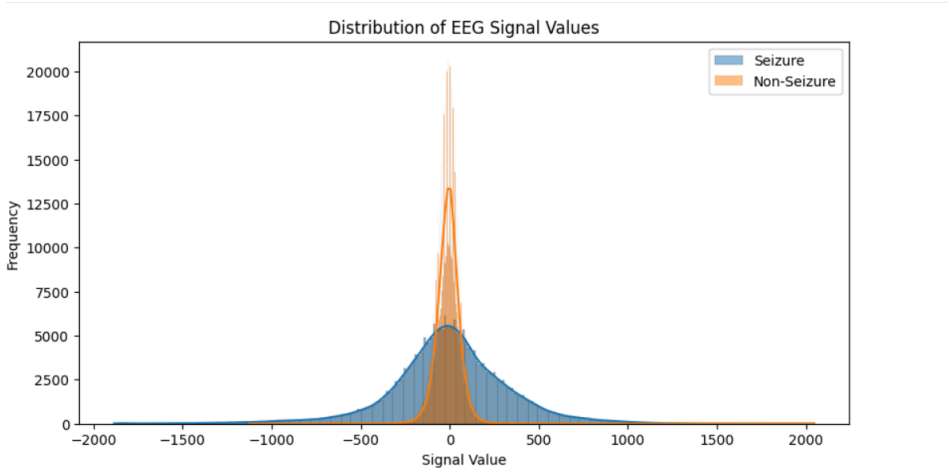
Exploratory data analysis was conducted to better understand the structure and characteristics of EEG signals and to identify patterns relevant to seizure detection. The class distribution analysis revealed a clear imbalance in the dataset after binary conversion, with approximately 2,300 seizure samples and a significantly larger number of non-seizure samples. This imbalance reflects real-world scenarios, where seizure events are relatively rare, and highlights the importance of considering class distribution during model training.



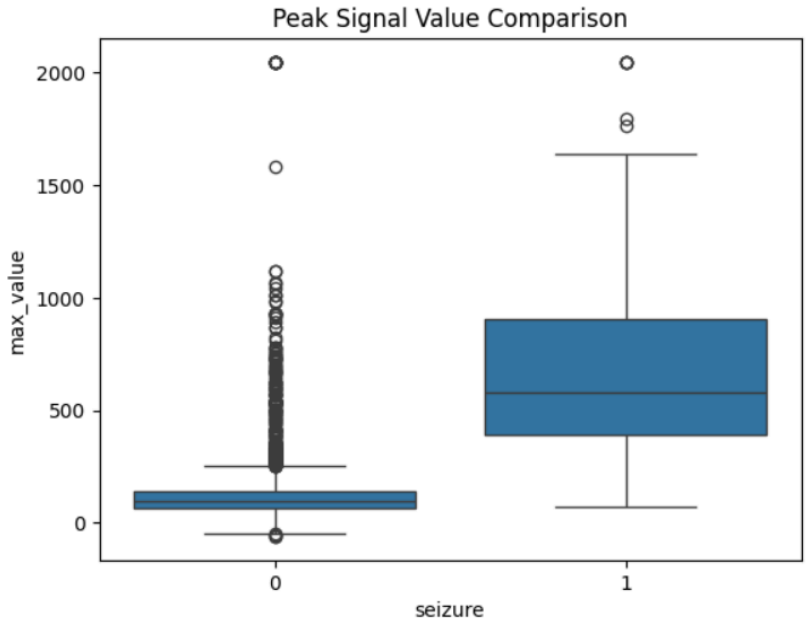
Further analysis involved comparing EEG signals between seizure and non-seizure samples. The results showed that seizure signals exhibit large spikes, rapid fluctuations, and high variability, whereas non-seizure signals appear smoother and more stable over time. This indicates that seizure activity produces distinct patterns in EEG signals that can be effectively captured by machine learning models.



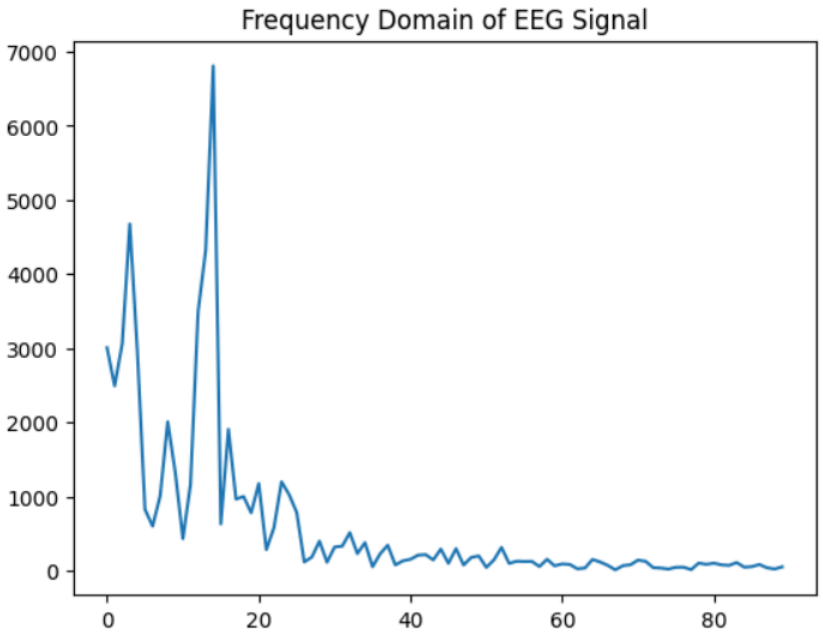
The distribution of signal values was also examined, showing that seizure signals have a wider spread and higher variance, while non-seizure signals are more concentrated around stable values. Statistical summaries of the dataset further confirmed the presence of high variability and extreme values across the 178 features, which is typical for EEG data. Despite this variability, the features demonstrated consistent scaling, making them suitable for analysis.



To further investigate signal characteristics, peak analysis was performed by extracting the maximum value from each EEG segment and visualizing it using a boxplot. This comparison was used to examine differences in peak amplitude between seizure and non-seizure EEG signals. The results showed that seizure signals generally have higher peak values and a wider spread than non-seizure signals, indicating stronger and more irregular signal behavior. This suggests that peak amplitude is an important feature for distinguishing seizure from non-seizure activity.



In addition, frequency domain analysis was conducted using the Fourier Transform to examine EEG signals beyond the time domain. This approach helps reveal spectral patterns that are not easily visible in raw signal plots. The frequency-domain visualization provides insight into how signal energy is distributed across different frequencies, highlighting important characteristics of EEG signals. This supports a deeper understanding of the dataset and its suitability for machine learning-based seizure classification.



Overall, these visualizations demonstrate that EEG data contains clear and distinguishable patterns, making it suitable for machine learning models to accurately classify seizure and non-seizure activity.

5. Results Summary

The evaluation results are summarized as follows:

- **Logistic Regression** achieved an accuracy of **0.71 (71%)**. However, it performed poorly in detecting seizure cases, with a seizure recall of **0.46**. This indicates that the model is biased toward the majority class (non-seizure) and struggles to capture the complex, non-linear patterns present in EEG signals.

- **Random Forest** achieved an accuracy of **0.974 (97.4%)**. It demonstrated strong seizure detection capability, with a recall of **0.91**, indicating high sensitivity to abnormal EEG patterns and effective handling of complex data relationships.
- **Support Vector Machine (SVM)** achieved the highest accuracy of **0.978 (97.8%)** and performed exceptionally well in seizure detection, with a recall of **0.98**. The model benefited from feature scaling and the use of a non-linear RBF kernel, allowing it to capture intricate patterns in the EEG signals.

Model	Accuracy	Seizure Recall	F1-score
Logistic Regression	0.71	0.46	0.39
Random Forest	0.97	0.91	0.93
SVM	0.98	0.98	0.95

5.1 Discussion

The results clearly show that non-linear and ensemble models outperform linear models for EEG-based seizure detection.

Logistic Regression fails to capture the complexity of EEG signals due to its linear nature, resulting in weak performance, particularly in detecting seizure cases.

In contrast, both Random Forest and SVM successfully model non-linear relationships and extract meaningful patterns from high-dimensional EEG data. Random Forest provides strong and stable performance, while SVM achieves the best overall results.

SVM achieved the highest performance across all evaluation metrics, particularly in seizure detection (recall = 0.98), making it the most effective model for this task.

5.2 Final Model Selection

Based on the evaluation results, the Support Vector Machine (SVM) model was selected as the final model due to:

- Highest accuracy (**97.8%**)
- Best seizure detection performance (recall = **0.98**)

- Strong generalization across both classes
- Effective handling of non-linear EEG patterns

6. Limitations

Despite strong performance results, this study has several limitations. The dataset used is not collected in real-time conditions, which may limit generalization to real-world wearable systems. Additionally, although class imbalance was present after binary conversion, no explicit resampling techniques were applied. Finally, only traditional machine learning models were explored, while deep learning approaches may further improve performance.

7. Conclusion

This study explored epileptic seizure detection using EEG signal data, with a strong emphasis on data exploration and understanding brain signal behavior.

The exploratory data analysis (EDA) played a central role in this project. Through statistical summaries, visualizations, and signal comparisons, clear differences between seizure and non-seizure EEG patterns were identified. Seizure signals exhibited higher variability, stronger fluctuations, and more extreme values compared to normal brain activity. These findings highlight the inherent complexity of EEG data and provide valuable insights into seizure characteristics.

Although machine learning models were applied for classification, the main contribution of this work lies in the analysis and interpretation of EEG signal patterns. The EDA phase established a strong foundation for understanding the dataset and justified the feasibility of automated seizure detection.

Among the tested models, the **SVM model achieved the best overall performance** and was selected as the final model due to its superior accuracy and seizure detection capability.

Overall, this project demonstrates that combining exploratory data analysis with machine learning enables effective detection of epileptic seizure patterns and supports the development of intelligent healthcare monitoring systems.

References

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